SQL Functions: Conversion Functions, and Date Functions.

OneCompiler Upgrade AI MySQL Run Save

queries.sql

```sql
CREATE TABLE employees (
    emp_id INT PRIMARY KEY,
    emp_name VARCHAR(50),
    salary DECIMAL(10,2),
    hire_date DATE,
    salary_text VARCHAR(10)
);

INSERT INTO employees VALUES
(1, 'Asha', 45000.50, '2022-03-15', '45000'),
(2, 'Ravi', 52000.75, '2021-11-01', '52000'),
(3, 'Meena', 38000.00, '2023-07-20', '38000');

SELECT CAST('123' AS UNSIGNED) AS text_to_int;
SELECT CAST(45000.50 AS CHAR) AS number_to_text;
SELECT CAST('2024-05-10' AS DATE) AS text_to_date;
SELECT emp_name, CAST(salary_text AS UNSIGNED) + 1000 AS new_salary FROM employees;

SELECT CONVERT('456', UNSIGNED) AS text_to_int;
SELECT CONVERT(salary, CHAR) AS salary_as_text FROM employees;
SELECT CONVERT('2024-05-10', DATE) AS text_to_date;

SELECT '10' + 5 AS result;

SELECT FORMAT(1234567.891, 2) AS formatted_number;
SELECT DATE_FORMAT('2024-05-10', '%d-%m-%Y') AS formatted_date;
SELECT STR_TO_DATE('10-05-2024', '%d-%m-%Y') AS text_to_date;

SELECT NOW() AS current_datetime;
SELECT CURDATE() AS today;
SELECT CURTIME() AS time_now;

SELECT emp_name, hire_date,
       YEAR(hire_date) AS yr,
       MONTH(hire_date) AS mon,
```

I/O AI Agent

STDIN

Input for the program (Optional)

Output:

```
+-------------+
| text_to_int |
+-------------+
|         123 |
+-------------+
+----------------+
| number_to_text |
+----------------+
| 45000.50       |
+----------------+
+--------------+
| text_to_date |
+--------------+
| 2024-05-10   |
+--------------+
+----------+------------+
| emp_name | new_salary |
+----------+------------+
| Asha     |      46000 |
| Ravi     |      53000 |
```

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OneCompiler Upgrade AI MySQL Run Save

queries.sql

```sql
INSERT INTO employees VALUES
(1, 'Asha', 45000.50, '2022-03-15', '45000'),
(2, 'Ravi', 52000.75, '2021-11-01', '52000'),
(3, 'Meena', 38000.00, '2023-07-20', '38000');

SELECT CAST('123' AS UNSIGNED) AS text_to_int;
SELECT CAST(45000.50 AS CHAR) AS number_to_text;
SELECT CAST('2024-05-10' AS DATE) AS text_to_date;
SELECT emp_name, CAST(salary_text AS UNSIGNED) + 1000 AS new_salary FROM employees;

SELECT CONVERT('456', UNSIGNED) AS text_to_int;
SELECT CONVERT(salary, CHAR) AS salary_as_text FROM employees;
SELECT CONVERT('2024-05-10', DATE) AS text_to_date;

SELECT '10' + 5 AS result;

SELECT FORMAT(1234567.891, 2) AS formatted_number;
SELECT DATE_FORMAT('2024-05-10', '%d-%m-%Y') AS formatted_date;
SELECT STR_TO_DATE('10-05-2024', '%d-%m-%Y') AS text_to_date;

SELECT NOW() AS current_datetime;
SELECT CURDATE() AS today;
SELECT CURTIME() AS time_now;

SELECT emp_name, hire_date,
       YEAR(hire_date) AS yr,
       MONTH(hire_date) AS mon,
       DAY(hire_date) AS dy,
       DAYNAME(hire_date) AS day_name,
       MONTHNAME(hire_date) AS month_name
FROM employees;

SELECT LAST_DAY('2024-02-10') AS last_day_of_month;
SELECT WEEK('2024-05-10') AS week_number;
SELECT DATE_FORMAT(NOW(), '%W, %d %M %Y') AS full_date;
```

I/O AI Agent 79 ms

STDIN

Input for the program (Optional)

Output:

```
+----------+------------+------+------+------+----------+------------+
| Asha     | 2022-03-15 | 2022 |    3 |   15 | Tuesday  | March      |
| Ravi     | 2021-11-01 | 2021 |   11 |    1 | Monday   | November   |
| Meena    | 2023-07-20 | 2023 |    7 |   20 | Thursday | July       |
+----------+------------+------+------+------+----------+------------+

+-------------------+
| last_day_of_month |
+-------------------+
| 2024-02-29        |
+-------------------+

+-------------+
| week_number |
+-------------+
|          18 |
+-------------+

+--------------------------+
| full_date                |
+--------------------------+
| Friday, 02 October 2026 |
+--------------------------+
```

Ready Light Wiki

SQL Operators: Demonstrate the use of di”erent types of operators in SQL with examples, including Arithmetic Operators, Logical Operators, Comparison Operators.

OneCompiler Upgrade AI MySQL Run Save C

queries.sql

```sql
DROP TABLE IF EXISTS staff;

CREATE TABLE staff (
    emp_id INT PRIMARY KEY,
    emp_name VARCHAR(50),
    salary DECIMAL(10,2),
    dept VARCHAR(20),
    age INT
);

INSERT INTO staff VALUES
(1, 'Asha',  45000, 'IT',      25),
(2, 'Ravi',  52000, 'HR',      30),
(3, 'Meena', 38000, 'IT',      28),
(4, 'Kiran', 60000, 'Finance', 35),
(5, 'Divya', 45000, NULL,      26);

SELECT * FROM staff;

SELECT 10 + 5  AS addition,
       10 - 5  AS subtraction,
       10 * 5  AS multiplication,
       10 / 4  AS division,
       10 DIV 4 AS integer_division,
       10 % 3  AS remainder;

       SELECT emp_name, salary,
       salary + 5000  AS after_raise,
       salary - 2000  AS after_deduction,
       salary * 0.10  AS bonus,
       salary / 12    AS monthly_salary,
       age % 2        AS age_remainder
FROM staff;

SELECT * FROM staff WHERE salary = 45000;
```

I/O AI Agent 83 ms

STDIN

Input for the program (Optional)

Output:

```
+--------+----------+----------+---------+------+
| emp_id | emp_name | salary   | dept    | age  |
+--------+----------+----------+---------+------+
|      1 | Asha     | 45000.00 | IT      |   25 |
|      2 | Ravi     | 52000.00 | HR      |   30 |
|      3 | Meena    | 38000.00 | IT      |   28 |
|      4 | Kiran    | 60000.00 | Finance |   35 |
|      5 | Divya    | 45000.00 | NULL    |   26 |
+--------+----------+----------+---------+------+
+----------+-------------+----------------+----------+------------------+-----------+
| addition | subtraction | multiplication | division | integer_division | remainder |
+----------+-------------+----------------+----------+------------------+-----------+
|       15 |           5 |             50 |   2.5000 |                2 |         1 |
+----------+-------------+----------------+----------+------------------+-----------+
+----------+----------+-------------+-----------------+-----------+-----------------+---------
| emp_name | salary   | after_raise | after_deduction | bonus     | monthly_salary | age_rema
+----------+----------+-------------+-----------------+-----------+-----------------+---------
| Asha     | 45000.00 |    50000.00 |        43000.00 | 4500.0000 |    3750.000000 |
| Ravi     | 52000.00 |    57000.00 |        50000.00 | 5200.0000 |    4333.333333 |
| Meena    | 38000.00 |    43000.00 |        36000.00 | 3800.0000 |    3166.666667 |
| Kiran    | 60000.00 |    65000.00 |        58000.00 | 6000.0000 |    5000.000000 |
| Divya    | 45000.00 |    50000.00 |        43000.00 | 4500.0000 |    3750.000000 |
```

Ready Light Wiki

OneCompiler Upgrade AI MySQL Run Save C

queries.sql

```
32        age % 2        AS age_remainder
33  FROM staff;
34
35  SELECT * FROM staff WHERE salary = 45000;
36  SELECT * FROM staff WHERE salary <> 45000;
37  SELECT * FROM staff WHERE salary > 45000;
38  SELECT * FROM staff WHERE salary < 45000;
39  SELECT * FROM staff WHERE salary >= 45000;
40  SELECT * FROM staff WHERE salary <= 45000;
41
42  SELECT * FROM staff WHERE salary BETWEEN 40000 AND 55000;
43  SELECT * FROM staff WHERE dept IN ('IT', 'HR');
44  SELECT * FROM staff WHERE emp_name LIKE 'A%';
45  SELECT * FROM staff WHERE emp_name LIKE '%a';
46  SELECT * FROM staff WHERE dept IS NULL;
47  SELECT * FROM staff WHERE dept IS NOT NULL;
48
49  SELECT NULL = NULL AS normal_equal, NULL <=> NULL AS null_safe_equal;
50
51  SELECT * FROM staff WHERE dept = 'IT' AND salary > 40000;
52
53  SELECT * FROM staff WHERE dept = 'IT' OR dept = 'HR';
54
55  SELECT * FROM staff WHERE NOT dept = 'IT';
56
57  SELECT * FROM staff WHERE dept = 'IT' XOR salary > 40000;
58
59  SELECT * FROM staff
60  WHERE (dept = 'IT' OR dept = 'HR') AND age > 26;
61
62  SELECT (1 AND 1) AS and_result,
63         (1 AND 0) AS and_false,
64         (1 OR 0)  AS or_result,
65         NOT 1     AS not_result,
66         (1 XOR 1) AS xor_result;
```

I/O AI Agent 83 ms

STDIN

Input for the program (Optional)

Output:

```
+--------+----------+----------+---------+------+
|      2 | Ravi     | 52000.00 | HR      |   30 |
|      4 | Kiran    | 60000.00 | Finance |   35 |
+--------+----------+----------+---------+------+
+--------+----------+----------+---------+------+
| emp_id | emp_name | salary   | dept    | age  |
+--------+----------+----------+---------+------+
|      2 | Ravi     | 52000.00 | HR      |   30 |
|      3 | Meena    | 38000.00 | IT      |   28 |
|      4 | Kiran    | 60000.00 | Finance |   35 |
+--------+----------+----------+---------+------+
+--------+----------+----------+------+------+
| emp_id | emp_name | salary   | dept | age  |
+--------+----------+----------+------+------+
|      2 | Ravi     | 52000.00 | HR   |   30 |
|      3 | Meena    | 38000.00 | IT   |   28 |
+--------+----------+----------+------+------+
+------------+-----------+-----------+------------+------------+
| and_result | and_false | or_result | not_result | xor_result |
+------------+-----------+-----------+------------+------------+
|          1 |         0 |         1 |          0 |          0 |
+------------+-----------+-----------+------------+------------+
```

SQL Operators: Demonstrate the use of di”erent types of
operators in SQL with examples Special Operators, and Set Operations(UNION)

OneCompiler Upgrade AI MySQL Run Save

queries.sql

```sql
DROP TABLE IF EXISTS staff;
CREATE TABLE staff (
    emp_id INT PRIMARY KEY,
    emp_name VARCHAR(50),
    salary DECIMAL(10,2),
    dept VARCHAR(20),
    age INT
);

INSERT INTO staff VALUES
(1, 'Asha',  45000, 'IT',      25),
(2, 'Ravi',  52000, 'HR',      30),
(3, 'Meena', 38000, 'IT',      28),
(4, 'Kiran', 60000, 'Finance', 35),
(5, 'Divya', 45000, NULL,      26);

DROP TABLE IF EXISTS contractors;
CREATE TABLE contractors (
    emp_id INT PRIMARY KEY,
    emp_name VARCHAR(50),
    salary DECIMAL(10,2),
    dept VARCHAR(20),
    age INT
);

INSERT INTO contractors VALUES
(101, 'Rohan', 40000, 'IT', 29),
(102, 'Asha',  45000, 'IT', 25),
(103, 'Sneha', 48000, 'HR', 27);

SELECT * FROM staff WHERE salary BETWEEN 40000 AND 55000;
SELECT * FROM staff WHERE salary NOT BETWEEN 40000 AND 55000;

SELECT * FROM staff WHERE dept IN ('IT', 'HR');
SELECT * FROM staff WHERE dept NOT IN ('IT', 'HR');
```

I/O AI Agent

STDIN

Input for the program (Optional)

Output:

```
|      1 | Asha     | 45000.00 | IT   |   25 |
|      2 | Ravi     | 52000.00 | HR   |   30 |
|      5 | Divya    | 45000.00 | NULL |   26 |
+--------+----------+----------+------+------+

+--------+----------+----------+---------+------+
| emp_id | emp_name | salary   | dept    | age  |
+--------+----------+----------+---------+------+
|      3 | Meena    | 38000.00 | IT      |   28 |
|      4 | Kiran    | 60000.00 | Finance |   35 |
+--------+----------+----------+---------+------+

+--------+----------+----------+------+------+
| emp_id | emp_name | salary   | dept | age  |
+--------+----------+----------+------+------+
|      1 | Asha     | 45000.00 | IT   |   25 |
|      2 | Ravi     | 52000.00 | HR   |   30 |
|      3 | Meena    | 38000.00 | IT   |   28 |
+--------+----------+----------+------+------+

+--------+----------+----------+---------+------+
| emp_id | emp_name | salary   | dept    | age  |
+--------+----------+----------+---------+------+
|      4 | Kiran    | 60000.00 | Finance |   35 |
+--------+----------+----------+---------+------+
```

Ready Light

OneCompiler Upgrade AI MySQL Run Save C

queries.sql

```sql
    dept VARCHAR(20),
    age INT
);

INSERT INTO contractors VALUES
(101, 'Rohan', 40000, 'IT', 29),
(102, 'Asha',  45000, 'IT', 25),
(103, 'Sneha', 48000, 'HR', 27);

SELECT * FROM staff WHERE salary BETWEEN 40000 AND 55000;
SELECT * FROM staff WHERE salary NOT BETWEEN 40000 AND 55000;

SELECT * FROM staff WHERE dept IN ('IT', 'HR');
SELECT * FROM staff WHERE dept NOT IN ('IT', 'HR');

SELECT * FROM staff WHERE emp_name LIKE 'A%';
SELECT * FROM staff WHERE emp_name LIKE '%a';
SELECT * FROM staff WHERE emp_name LIKE '%i%';
SELECT * FROM staff WHERE emp_name LIKE '_a%';

SELECT * FROM staff WHERE dept IS NULL;
SELECT * FROM staff WHERE dept IS NOT NULL;

SELECT emp_name FROM staff s
WHERE EXISTS (SELECT 1 FROM contractors c WHERE c.emp_name = s.emp_name);

SELECT emp_name, salary FROM staff
WHERE salary > ALL (SELECT salary FROM contractors WHERE dept = 'IT');

SELECT emp_name, salary FROM staff
WHERE salary > ANY (SELECT salary FROM contractors WHERE dept = 'IT');

SELECT emp_name, dept FROM staff
UNION
SELECT emp_name, dept FROM contractors;
```

I/O AI Agent 93 ms

STDIN

Input for the program (Optional)

Output:

```
| Ravi     | 52000.00 |
| Kiran    | 60000.00 |
+----------+----------+

+----------+----------+
| emp_name | salary   |
+----------+----------+
| Asha     | 45000.00 |
| Ravi     | 52000.00 |
| Kiran    | 60000.00 |
| Divya    | 45000.00 |
+----------+----------+

+----------+---------+
| emp_name | dept    |
+----------+---------+
| Asha     | IT      |
| Ravi     | HR      |
| Meena    | IT      |
| Kiran    | Finance |
| Divya    | NULL    |
| Rohan    | IT      |
| Sneha    | HR      |
+----------+---------+
```